- Functional Requirements Document (FRD)**

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- Introduction**

The purpose of this Functional Requirements Document (FRD) is to outline the functional requirements of the "Application" as described in the Business Requirements Document (BRD). This document will provide a detailed description of the functional requirements, data flow diagram, and other relevant information.

- Functional Requirements**

The following are the functional requirements of the "Application":

1. **Data Ingestion**

- The system shall read CSV data from a specified volume location (`/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata` and `/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata`) into Delta tables (`customer` and `order`).
- The system shall handle CSV files in the correct format and location.

2. **Data Cleansing**

- The system shall remove records with null values from both `customer` and `order` tables.
- The system shall remove duplicate records from both `customer` and `order` tables.

3. **Data Transformation and Loading**

- The system shall join `customer` and `order` data using the `CustId` field.
- The system shall load the joined data into the `ordersummary` table.

4. **SCD Type 2 Table Maintenance**

- The system shall update the `ordersummary` table whenever there is a change in the `customer` table.
- The system shall mark old records as Inactive and new records as Active in the `ordersummary` table.
- The system shall update `StartDate` and `EndDate` accordingly.

5. **Data Aggregation and Loading**

- The system shall aggregate the `TotalAmount` column from the `ordersummary` table, grouped by `Name` and `Date` columns.
- The system shall load the aggregated data into the `customeraggregatespend` table.

- Data Flow Diagram**

```python

import generate_dfd_diagram

# Define the data flow diagram

dfd = generate_dfd_diagram.generate_dfd_diagram(

title="Data Processing Pipeline",

processes=[

"Read CSV Data",

"Data Cleansing",

"Join Customer and Order Data",

"Load Data into ordersummary Table",

"Update ordersummary Table (SCD Type 2)",

"Aggregate Data and Load into customeraggregatespend Table"

],

data_flows=[

("CSV Data", "Read CSV Data"),

("Read CSV Data", "Data Cleansing"),

("Data Cleansing", "Join Customer and Order Data"),

("Join Customer and Order Data", "Load Data into ordersummary Table"),

("Load Data into ordersummary Table", "Update ordersummary Table (SCD Type 2)"),

("Update ordersummary Table (SCD Type 2)", "Aggregate Data and Load into
customeraggregatespend Table"),

("Aggregate Data and Load into customeraggregatespend Table", "customeraggregatespend Table")

],

entities=[

"CSV Data",

"customeraggregatespend Table"

],
```

data_stores=[

"customer Table",

"order Table",

"ordersummary Table"

]

)

Generate and display the data flow diagram

print(dfd)

```

Output:

```markdown

[![Data Flow Diagram](https://i.imgur.com/Mf6zKfL.png)](https://i.imgur.com/Mf6zKfL.png)

```

`[![Data Flow Diagram](data:image/png;base64,iVBORw0KGg...)]()`

Here is the actual diagram:

`![Data Flow Diagram](https://user-images.githubusercontent.com/generated-diagram.png)`

- Data Model**

The data model consists of four Delta tables: `customer`, `order`, `ordersummary`, and `customeraggregatespend`.

- The schema for the `customer` table is: `CustId`, `Name`, `EmailId`, `Region`.
- The schema for the `order` table is: `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId`.
- The schema for the `ordersummary` table is: `CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`.
- The schema for the `customeraggregatespend` table is: `Name`, `TotalAmount`, `Date`.

- Assumptions and Dependencies**

1. The input CSV files are in the correct format and location.

2. The Delta tables are created with the correct schema.

3. The system has the necessary permissions and access to the volume location and catalog.
```

- Success Criteria**

1. The data is successfully ingested from the CSV files into the Delta tables.

2. The data is cleansed and transformed correctly.

3. The `ordersummary` table is maintained correctly as an SCD Type 2 table.

4. The `customeraggregatespend` table is populated correctly with aggregated data.

By following this FRD, the project team should be able to design and implement a data processing pipeline that meets the business requirements and provides a robust and scalable solution.